AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 25 · STUDENT LESSON

Sequence representations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Keep the order. Mark real positions. Preserve measured zero.

Encode complete sequence schemas while tracing padding,unknown identity and truncation losses.

Keep one habit: Order collision → four-position vocabulary/encoding → separate pictured sensor mask → aligned real-zero diagram → executable mask audit → independent new schema.

MAIN LESSON · SECTION 1

Which event was last,and which positions are real?

> Keep order,vocabulary,storage length and masks together.

Learning goals

I can encode an ordered **sequence** with an explicit vocabulary.

I can use **padding** and a **mask** without inventing real events.

I can identify information lost by counts unknown symbols and truncation.

Retrieve representation and evidence limits

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Why can two different image regions have the same pooled maximum?

> **My starting explanation**

End goal

By the end, I can explain **sequence** representations using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Encode an ordered **sequence** with an explicit vocabulary

Use **padding** and a **mask** without inventing real events

Identify information lost by counts unknown symbols and truncation

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Choose a representation that keeps the order this task needs

> Counts alone can merge logs with different final events.

Complete event task and order collision

Use fictional robot event logs whose symbols are GO, TURN and STOP. The task asks whether the final real event is STOP. This is a representation exercise, not a claim that an event log proves a robot is safe.

GO then STOP and STOP then GO contain the same symbols but have different final events.

Log	Counts in GO TURN STOP order	Final event	Task answer
GO STOP	[1, 0, 1]	STOP	Yes
STOP GO	[1, 0, 1]	GO	No
GO TURN STOP	[1, 1, 1]	STOP	Yes

Declared vocabulary and the meaning of IDs

A count vector discards order, so no deterministic rule using only the first two count vectors can distinguish their answers. A **sequence** of IDs preserves order if we retain each position.

Declare the vocabulary: PAD = 0, UNK = 1, GO = 2, TURN = 3 and STOP = 4. PAD is reserved for added storage positions; UNK represents a real but unrecognised symbol.

IDs are names, not measured amounts. STOP having ID 4 does not make it twice as much as GO with ID 2. Changing the vocabulary requires changing every consumer consistently.

Do not interpret numerical distance between IDs as semantic similarity.

Name given and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Encode real positions before adding storage padding

> A token ID names an event;the mask records whether a position is real.

Complete input contract and training-data vocabulary scope

Store each log in four positions, **padding** on the right. A **mask** value 1 marks a real position and 0 marks **padding**. The encoder rejects empty logs, logs longer than four, and literal reserved markers.

Other unrecognised symbols become UNK and still receive **mask** 1.

The code assumes a list of symbol strings and uses exact spelling. It does not fit a vocabulary from data.

If a future project fits vocabulary choices or frequency cutoffs, derive those choices from **training** data and apply the saved encoding consistently to validation and final cases.

All supplied encodings and last real positions

Complete input log	Four stored IDs	Real-position mask	Final real position / meaning
GO STOP	[2, 4, 0, 0]	[1, 1, 0, 0]	1 /STOP
STOP GO	[4, 2, 0, 0]	[1, 1, 0, 0]	1 /GO
TURN STOP	[3, 4, 0, 0]	[1, 1, 0, 0]	1 /STOP
WAIT STOP	[1, 4, 0, 0]	[1, 1, 0, 0]	1 /STOP; WAITidentity lost at real position 0
GO TURN GO STOP	[2, 3, 2, 4]	[1, 1, 1, 1]	3 /STOP; no padding

Vocabulary,unknown and invalid-input distinctions

The encoder uses PAD0, UNK1, GO2, TURN3, STOP4 and exactly four stored positions with right padding. Unknown actual symbol strings consume a real position and get mask 1. Missing input is a different problem:the function assumes a list of actual strings and does not validate every type. Literal reserved marker strings PAD/UNK, empty logs and more than four real events are rejected.

The appended visible examples print IDs and masks; they do not train a network or infer event safety.

Runnable encoder with visible worked outputs

```
VOCAB = {"GO": 2, "TURN": 3, "STOP": 4}

def encode(events):
    if not 1 <= len(events) <= 4:
        raise ValueError("Need one to four real events")
    if any(e in ("PAD", "UNK") for e in events):
        raise ValueError("Reserved markers are not events")
    ids = [VOCAB.get(e, 1) for e in events]
    n = len(ids)
    return ids + [0] * (4 - n), [1] * n + [0] * (4 - n)
```

```
assert encode(["GO", "STOP"]) == ([2, 4, 0, 0], [1, 1, 0, 0])
assert encode(["WAIT", "STOP"]) == ([1, 4, 0, 0], [1, 1, 0, 0])
assert encode(["STOP", "GO"]) == ([4, 2, 0, 0], [1, 1, 0, 0])
for events in [['GO','STOP'],['STOP','GO'],['WAIT','STOP']]:
    ids,mask=encode(events)
    print(events,'IDs',ids,'mask',mask,'real length',sum(mask))
```

Connect the storage diagram to IDs and mask positions

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Reference: original assertion-only encoder

```
VOCAB = {"GO": 2, "TURN": 3, "STOP": 4}

def encode(events):
    if not 1 <= len(events) <= 4:
        raise ValueError("Need one to four real events")
    if any(e in ("PAD", "UNK") for e in events):
        raise ValueError("Reserved markers are not events")
    ids = [VOCAB.get(e, 1) for e in events]
    n = len(ids)
    return ids + [0] * (4 - n), [1] * n + [0] * (4 - n)

assert encode(["GO", "STOP"]) == ([2, 4, 0, 0], [1, 1, 0, 0])
assert encode(["WAIT", "STOP"]) == ([1, 4, 0, 0], [1, 1, 0, 0])
assert encode(["STOP", "GO"]) == ([4, 2, 0, 0], [1, 1, 0, 0])
```

▼ Earlier wording, alternatives and picture transcripts

Connect the picture and the explanation

Change order without changing the token counts

> Trace which distinction the downstream question requires.

Current three-event count-collision task

- Give two logs made of exactly GO, TURN and STOP once each that have equal count vectors but different answers to whether the final event is STOP.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured value-only padding decision

- Change: A missing **sequence** length causes every numeric 0 to be treated as **padding** in the sensor example.

▼ Check after predicting

Prediction: The first real sensor reading is incorrectly discarded.

Reason: Its numeric value is 0, but its position is real according to the declared length.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Use length or mask to distinguish a real zero from padding

> Token identifiers and sensor measurements use different value meanings.

Complete pictured vocabulary,sensor values and scope

Distinguish **padding** positions from meaningful zero sensor readings.

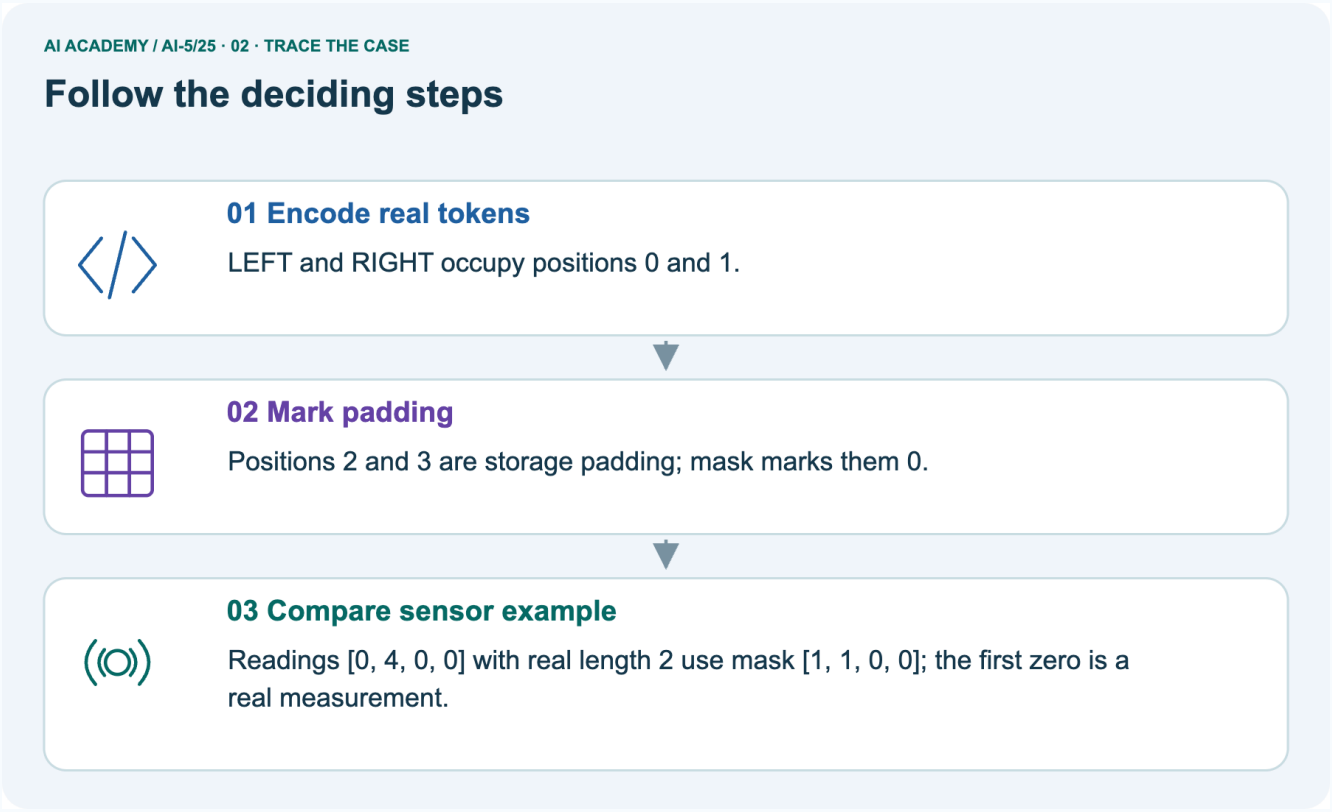
Toy tokens PAD 0, UNK 1, LEFT 2, RIGHT 3; stored length 4. **Sequence** LEFT RIGHT encodes [2, 3, 0, 0], **mask** [1, 1, 0, 0].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses a separate vocabulary:PAD0, UNK1, LEFT2, RIGHT3 and four stored positions. LEFT RIGHTgives IDs[2, 3, 0, 0], mask[1, 1, 0, 0]. Its sensor example[0, 4, 0, 0]uses the same mask because the first two measurements are real. The first zero is measured; the last two are storage placeholders. Removing every numeric zero would wrongly discard that measurement.

Keep this LEFT/RIGHTvocabulary separate from the main GO/TURN/STOPvocabulary; the same IDnumber has meaning only under its declared mapping.

Trace real and placeholder positions



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Encode real tokens	LEFT and RIGHT occupy positions 0 and 1.
2. Mark padding	Positions 2 and 3 are storage padding; mask marks them 0.
3. Compare sensor example	Readings [0, 4, 0, 0] with real length 2 use mask [1, 1, 0, 0]; the first zero is a real measurement.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

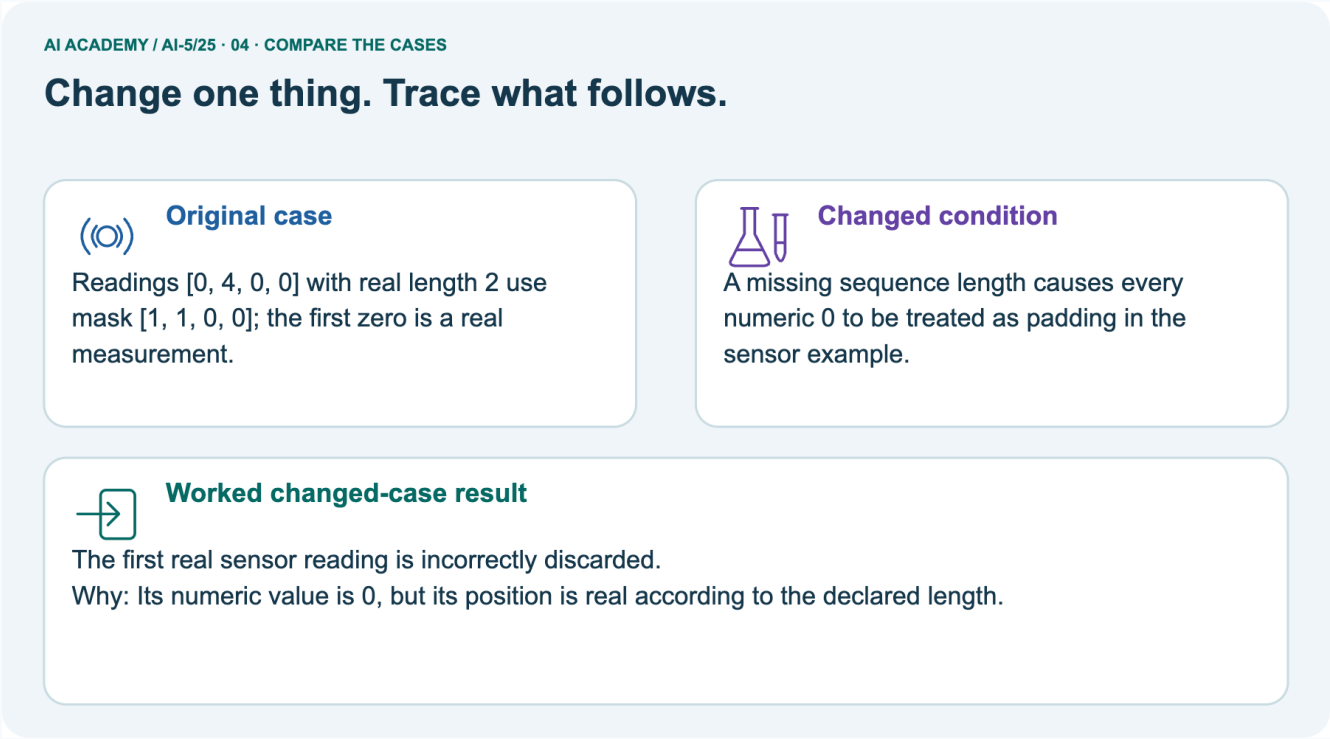
Sequence values and **sequence** length are separate information. **Padding** fills unused storage positions so arrays can share a size; a **mask** marks which positions are real.

A real sensor reading can be zero, just like a **padding** value. Keep the length or **mask** rather than inferring presence from the number alone. **Token ID**s are identifiers, not measured magnitudes.

Predict the missing-length error

- Change one thing: A missing **sequence** length causes every numeric 0 to be treated as **padding** in the sensor example. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the needed record and schema limit

- Token **padding** rules need not apply to sensor values. Document **sequence** lengths, vocabulary and unknown-token handling.

- Read the labelled picture: Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to main TURN STOPand its last real position

A padded log still has two real events

STORED POSITION	TOKEN ID	MASK
0	2 meaning GO	1 real
1	4 meaning STOP	1 real
2	0 meaning PAD	0 placeholder
3	0 meaning PAD	0 placeholder

The final stored slot is padding. For this right-padded contiguous mask, the real length is the mask sum, and the final real event is at length minus one. A different masking layout would need a different indexing rule.

Encode TURN STOP, give its mask and real length, then identify the final real position and its ID. Why would reading stored position 3 give the wrong event?

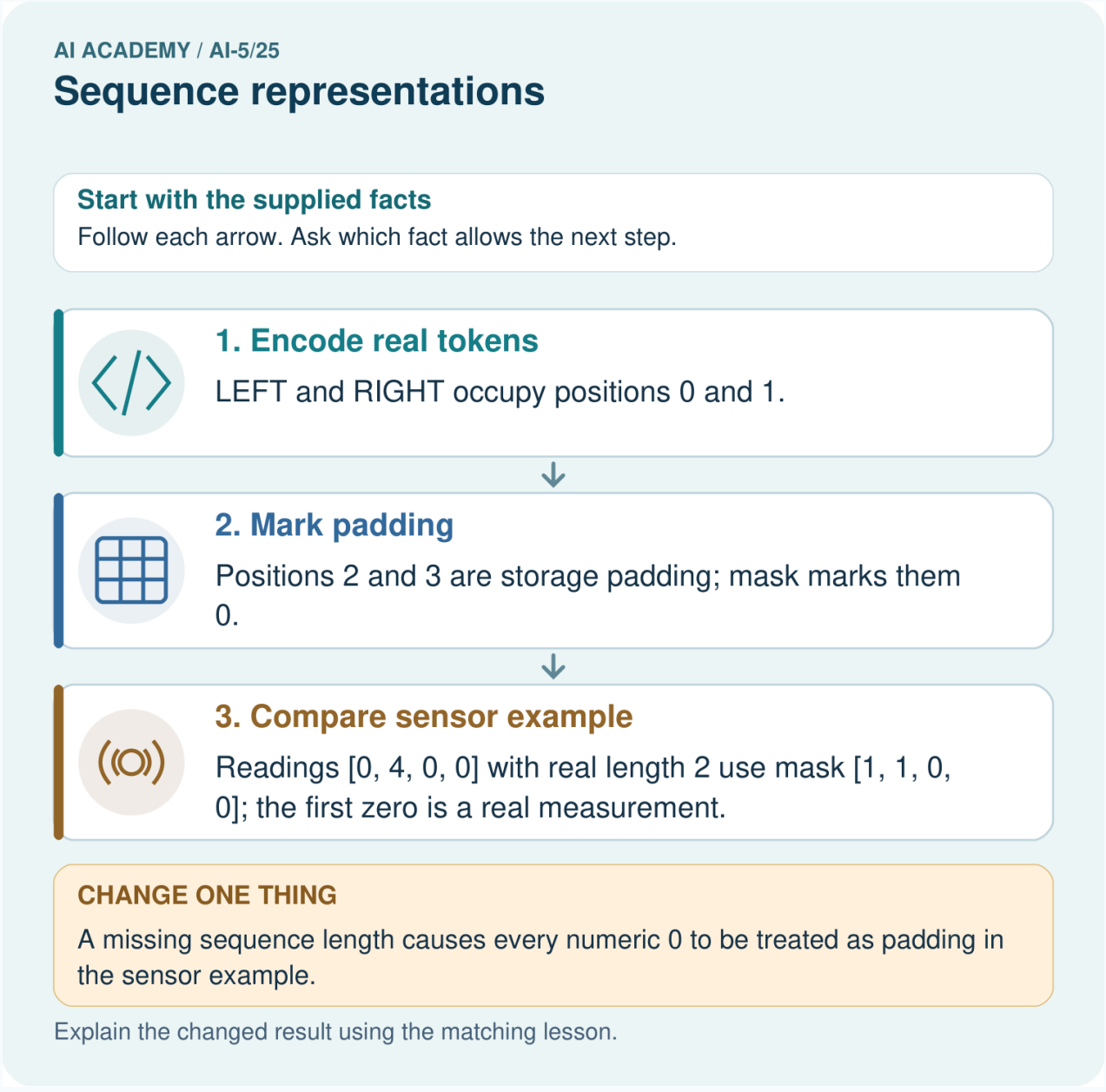
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sequence representations

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Make the mask affect the calculation

> Preserving ordered input also requires a later computation that uses the needed order.

Complete means,real-zero and downstream-order explanation

A **mask** must affect a calculation to protect it from **padding**. Suppose real sensor values are [2, 4], stored as [2, 4, 0, 0] with **mask** [1, 1, 0, 0]. The **mask**ed mean is (2 + 4) / 2 = 3. Averaging all four slots gives 1.5.

Unlike arbitrary **token ID**s, these sensor values are measurements for which a mean is meaningful.

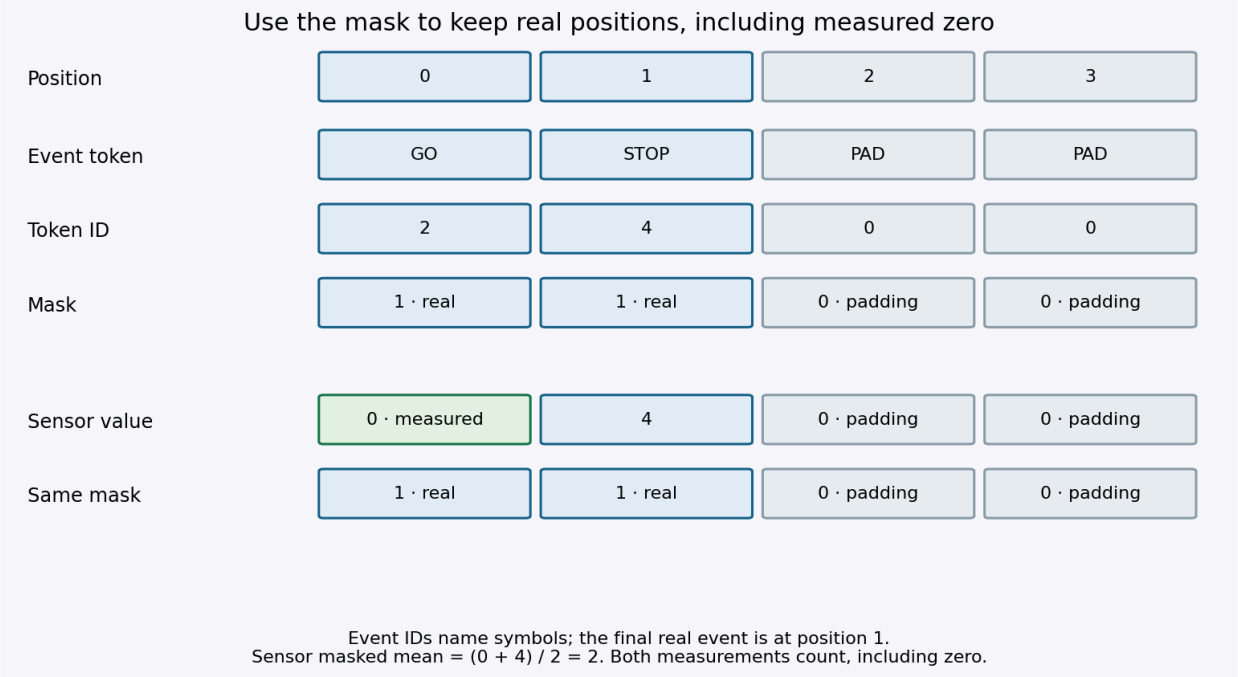
Measured zero must remain real: values [0, 4, 0, 0] with **mask** [1, 1, 0, 0] have mean 2. Removing every zero would incorrectly give 4. The **mask** distinguishes a measured zero from a storage placeholder.

An all-zero **mask** has no real values; reject it or follow an explicitly defined empty-**input** policy instead of dividing by zero.

Storing ordered IDs makes order available but does not guarantee a later model uses it. A computation that sums token counts or averages position-independent vectors can lose order again.

Trace what the whole procedure reads, not just what the **input** format could preserve.

Aligned event and measurement positions



Every correct and incorrect sensor denominator

Complete sensor calculation	Selected values / denominator	Result
Stored[2, 4, 0, 0], mask[1, 1, 0, 0]	Real[2, 4], divide by2	3
Wrong all-four-slot average	(2+4+0+0)/4	1.5
Stored[0, 4, 0, 0], same mask	Real[0, 4], divide by2	2
Wrong discard-every-zero policy	Only[4], divide by1	4
Wrong all-four-slot average for[0, 4, 0, 0]	4/4	1

Complete sensor calculation	Selected values / denominator	Result
All-zero mask	No real measurements	Reject or use an explicitly declared empty policy

Try the WAIT/PAUSEidentity collision

- Encode WAIT GO. Which positions are real? Compare it with PAUSE GO, and state what information the encoding loses.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise order and masks with less help

> Name the last real position from the mask rather than the final array slot.

Current task and response

- Encode GO TURN STOP using the lesson vocabulary and four positions. Give its **mask**, real length and final real event.

Encode TURN STOP, give its mask and real length, then identify the final real position and its ID. Why would reading stored position 3 give the wrong event?

Explain your prediction to a partner before comparing results.

Supported W2 record

Use workbook W2 for the connected example "**Sequence** representations". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Sequence** representations. Proposed explanation: “Larger **token ID**s mean more of a concept”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured real-zero reasoning

- Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent new vocabulary and measurements

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Expose an order collision

Check your reasoning

Explain the picture

Run a complete order,padding and measured-zero audit

> All vocabularies,event lists,masks,measurement values and invalid cases are supplied.

Current order-collision task

Give two logs made of exactly GO, TURN and STOP once each that have equal count vectors but different answers to whether the final event is STOP.

Complete runnable event and sensor audit

```
# Complete supplied-vocabulary encoder and separate sensor-mask investigation.
VOCAB = {'GO':2,'TURN':3,'STOP':4}

def encode(events):
    if not 1<=len(events)<=4:
        raise ValueError('Need one to four real events')
    if any(e in ('PAD','UNK') for e in events):
        raise ValueError('Reserved markers are not events')
    ids=[VOCAB.get(e,1) for e in events]
    return ids+[0]*(4-len(ids)),[1]*len(ids)+[0]*(4-len(ids))

def last_event(ids,mask):
    if len(ids)!=4 or len(mask)!=4 or any(m not in (0,1) for m in mask):
        raise ValueError('Need four IDs and a binary right-padding mask')
    n=sum(mask)
    if n==0 or mask!=[1]*n+[0]*(4-n):
        raise ValueError('Need real events followed by padding')
    if any(i not in (1,2,3,4) for i in ids[:n]) or any(i!=0 for i in ids[n:]):
        raise ValueError('Real IDs and padding IDs violate the schema')
    token_id=ids[n-1]
    # This demonstration declares review when the final event identity is unknown.
    answer='review' if token_id==1 else 'yes' if token_id==4 else 'no'
    return n-1,token_id,answer

def masked_mean(values,mask):
    if not values or len(values)!=len(mask) or any(m not in (0,1) for m in mask):
        raise ValueError('Need aligned measurements and a binary mask')
    real=[v for v,m in zip(values,mask) if m]
    if not real:
        raise ValueError('No real measurements')
    return sum(real)/len(real)
```

```
for events in [['GO','STOP'],['STOP','GO'],['WAIT','GO'],['PAUSE','GO'],['GO','WAIT']]:
    ids,mask=encode(events)
    print(events,'IDs',ids,'mask',mask,'final position/ID/result',last_event(ids,mask))
values,mask=[0,4,0,0],[1,1,0,0]
print('sensor masked mean',masked_mean(values,mask))
print('incorrect all-slot mean',sum(values)/len(values))
wrong_real=[v for v in values if v!=0]
print('incorrect discard-zero mean',sum(wrong_real)/len(wrong_real))
print('padding-only numeric changes leave masked mean',masked_mean([0,4,999,-7],mask))
print('count collision',['GO','TURN','STOP'],['STOP','TURN','GO'],'both counts',[1,1,1])
# Assume actual event strings and numeric sensor measurements. Missing input needs review.
# Token IDs are names; this program never treats their average as a measurement.
```

Normal,boundary,unknown,missing and truncation probes

Complete sequence probe	Expected result / action
Normal GO STOP	IDs[2, 4, 0, 0], mask[1, 1, 0, 0], last real position 1, STOP
Reversed STOP GO	Same counts[1, 0, 1], different final eventGO
Full four-event GO TURN GO STOP	Mask[1, 1, 1, 1], last position 3, STOP
Unknown WAIT GOand PAUSE GO	Both[1, 2, 0, 0], same real mask; unknown identity lost, but known lastGOremains available
GO WAITwith unknown final ID	The extended audit explicitly returns review under its declared unknown-final-event policy
Empty log or five real events	ValueError:Need one to four real events
Literal PAD/UNKmarker string in input	ValueError:Reserved markers are not events
Missing event value instead of an actual symbol string	Outside the declared type contract; request the event rather than invent one
Measured zero with correct sensor mask	Mean[0, 4]is2; wrong zero-filter mean 4 and all-slot mean 1
All-zero sensor mask or misaligned lengths	Extended masked_meanrejects invalid/no-real-measurement input

Complete sequence probe	Expected result / action
Truncate GO TURN GO TURN STOPto its first 4	Changes finalSTOPtoTURN; original encoder rejects the long log instead of silently changing the task

Predict,encode,mask and explain the actual result

1. Predict every complete encoding and sensor mean before running or tracing the encoder and extended audit. The extended last-event function validates the four-position/right-padding contract and declares review when its final real ID is unknown. The sensor mean uses only mask 1 positions and still counts a measured zero. It assumes actual numeric measurements, not arbitrary token IDs.

   Try changed numeric padding values to check whether the mask is actually used; keep the declared storage policy in the record. Preserve unknown/truncation losses and reject unsupported lengths rather than fabricate missing events.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

<details>
<summary>Reference: original compact encoder</summary>

```
VOCAB = {"GO": 2, "TURN": 3, "STOP": 4}

def encode(events):
    if not 1 <= len(events) <= 4:
        raise ValueError("Need one to four real events")
    if any(e in ("PAD", "UNK") for e in events):
        raise ValueError("Reserved markers are not events")
    ids = [VOCAB.get(e, 1) for e in events]
    n = len(ids)
    return ids + [0] * (4 - n), [1] * n + [0] * (4 - n)

assert encode(["GO", "STOP"]) == ([2, 4, 0, 0], [1, 1, 0, 0])
assert encode(["WAIT", "STOP"]) == ([1, 4, 0, 0], [1, 1, 0, 0])
assert encode(["STOP", "GO"]) == ([4, 2, 0, 0], [1, 1, 0, 0])
```

</details>

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Predict, investigate and preserve the evidence

Plan → try → check

</details>

MAIN LESSON · SECTION 14

Try alone: a new vocabulary,length and sensor sequence

> Use its five stored positions and retain its real measured zero.

Complete independent schema and measurements

Use PAD = 0, UNK = 1, OPEN = 2, CLOSE = 3, CHECK = 4 and five stored positions. Encode OPEN CHECK CLOSE and give its **mask** and final real position. Separately, sensor values [0, 6, 3, 0, 0] have that same **mask**.

Calculate their **mask**ed mean and the incorrect mean over all five slots.

First attempt and support record

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a fresh schema independently

Attempt this without hints

Use PAD = 0, UNK = 1, OPEN = 2, CLOSE = 3, CHECK = 4 and five stored positions. Encode OPEN CHECK CLOSE and give its **mask** and final real position. Separately, sensor values [0, 6, 3, 0, 0] have that same **mask**.

Calculate their **mask**ed mean and the incorrect mean over all five slots.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep sequence schema and lost-information evidence together

> The project record should identify vocabulary,real length,padding,mask and unknown/truncation policy.

Project artifact and evidence table

Add a labelled **sequence** representations **evidence** sheet to Image-pattern Network Investigation. Show how you can encode an ordered **sequence** with an explicit vocabulary.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and remaining limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one real-position trace and one lost distinction

> Explain why an ordered format alone cannot force a later model to use order.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Encode an ordered **sequence** with an explicit vocabulary

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Sequence	An ordered list of elements in which positions can matter
Token ID	A numerical identifier assigned to a symbol by a vocabulary
Padding	Placeholder positions added to make stored sequences the same length
Mask	A list marking which stored positions are real inputs and which are placeholders

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Encode GO TURN STOP using the lesson vocabulary and four positions. Give its **mask**, real length and final real event.
- Encode WAIT GO. Which positions are real? Compare it with PAUSE GO, and state what information the encoding loses.
- The encoder receives five events ending in STOP, then an empty log. What should it do under the lesson contract? Why should a program not simply crop the first and assume a final event for the second?

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent five-position vocabulary and measured-zero case for the first individual attempt.

Current explanation and response

- WAIT STOP and PAUSE STOP both become [1, 4, 0, 0] with the same **mask**. UNK retains a real position but loses the identity of the unrecognised symbol.

  The final-event task can still answer Yes for both because the last real ID is STOP. A final real UNK would require an unknown result or another documented policy when the intended event cannot be determined.

- The full log ends in STOP, while the retained prefix ends in TURN. Truncation changes the **evidence** needed for this task. Save the original length and any truncation policy in a real experiment, and test the con**sequence**s.

  Next week introduces embeddings while preserving the distinction between identifiers and learned vectors.

- The encoder receives five events ending in STOP, then an empty log. What should it do under the lesson contract? Why should a program not simply crop the first and assume a final event for the second?

> **A log GO TURN GO TURN STOP is cut to its first four events before encoding. What happens to the final-event answer? Explain why the lesson encoder rejects the original log instead of silently truncating it.**

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

A padded log still has two real events

STORED POSITION	TOKEN ID	MASK
0	2 meaning GO	1 real
1	4 meaning STOP	1 real
2	0 meaning PAD	0 placeholder
3	0 meaning PAD	0 placeholder

The final stored slot is padding. For this right-padded contiguous mask, the real length is the mask sum, and the final real event is at length minus one. A different masking layout would need a different indexing rule.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Distinguish **padding** positions from meaningful zero sensor readings.

By the end, I can	How I will show it
Encode an ordered sequence with an explicit vocabulary	Explain the deciding step, show a trace or test, and state one limit.
Use padding and a mask without inventing real events	Explain the deciding step, show a trace or test, and state one limit.
Identify information lost by counts unknown symbols and truncation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Distinguish **padding** positions from meaningful zero sensor readings.
- Which labelled step matches this detail? Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.
- 1. Encode real tokens
- 2. Compare sensor example
- 3. Mark **padding**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare sensor example Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.

Sequence values and **sequence** length are separate information. **Padding** fills unused storage positions so arrays can share a size; a **mask** marks which positions are real.

A real sensor reading can be zero, just like a **padding** value. Keep the length or **mask** rather than inferring presence from the number alone. **Token ID**s are identifiers, not measured magnitudes.

Changed-case question and optional hint

- Changed-case practice: A missing **sequence** length causes every numeric 0 to be treated as **padding** in the sensor example. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The first real sensor reading is incorrectly discarded.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The first real sensor reading is incorrectly discarded. Its numeric value is 0, but its position is real according to the declared length.

Repair and retain the record

Review Augmentation and leakage (ai-5/24). Goal: Check whether a transformation preserves the task label. Short practice: How many independent source groups are in the final test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learned representations (ai-5/19). Goal: Trace an **input** through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

</details>

<details>
<summary>Reference · Six-picture concept journey</summary>

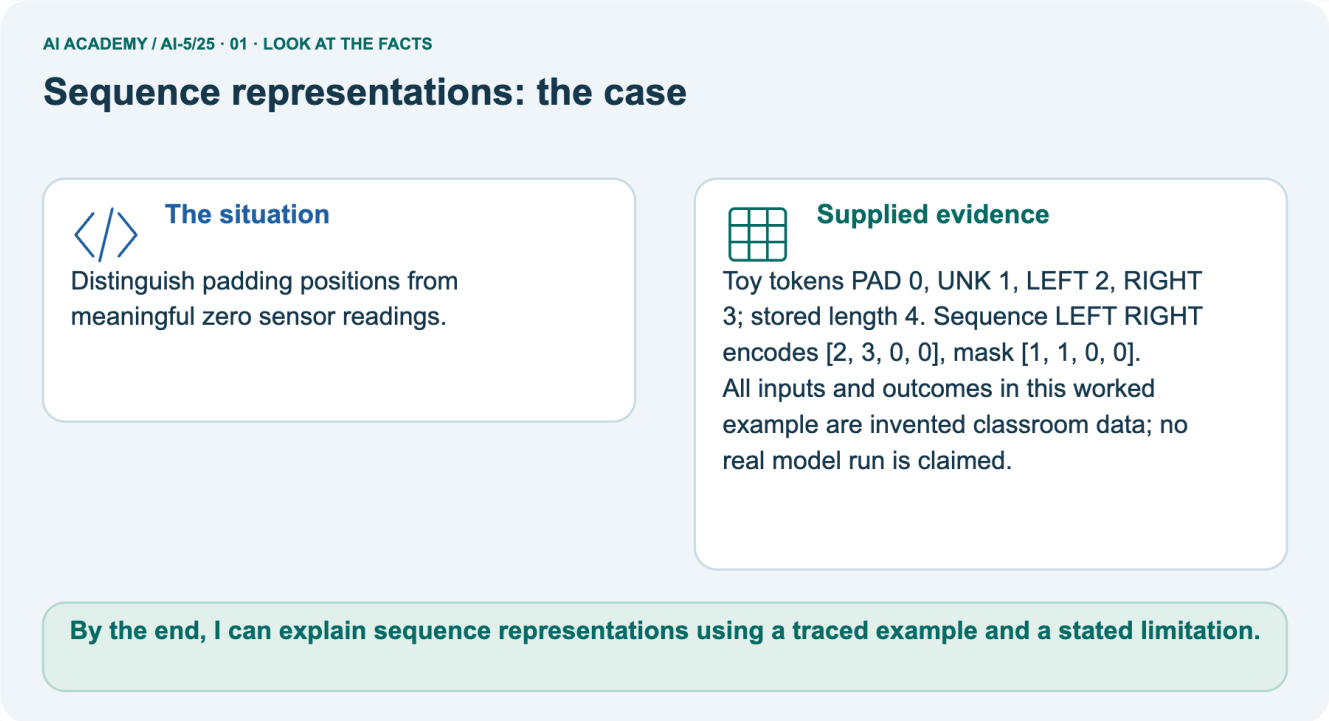
</details>

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Sequence representations: the case



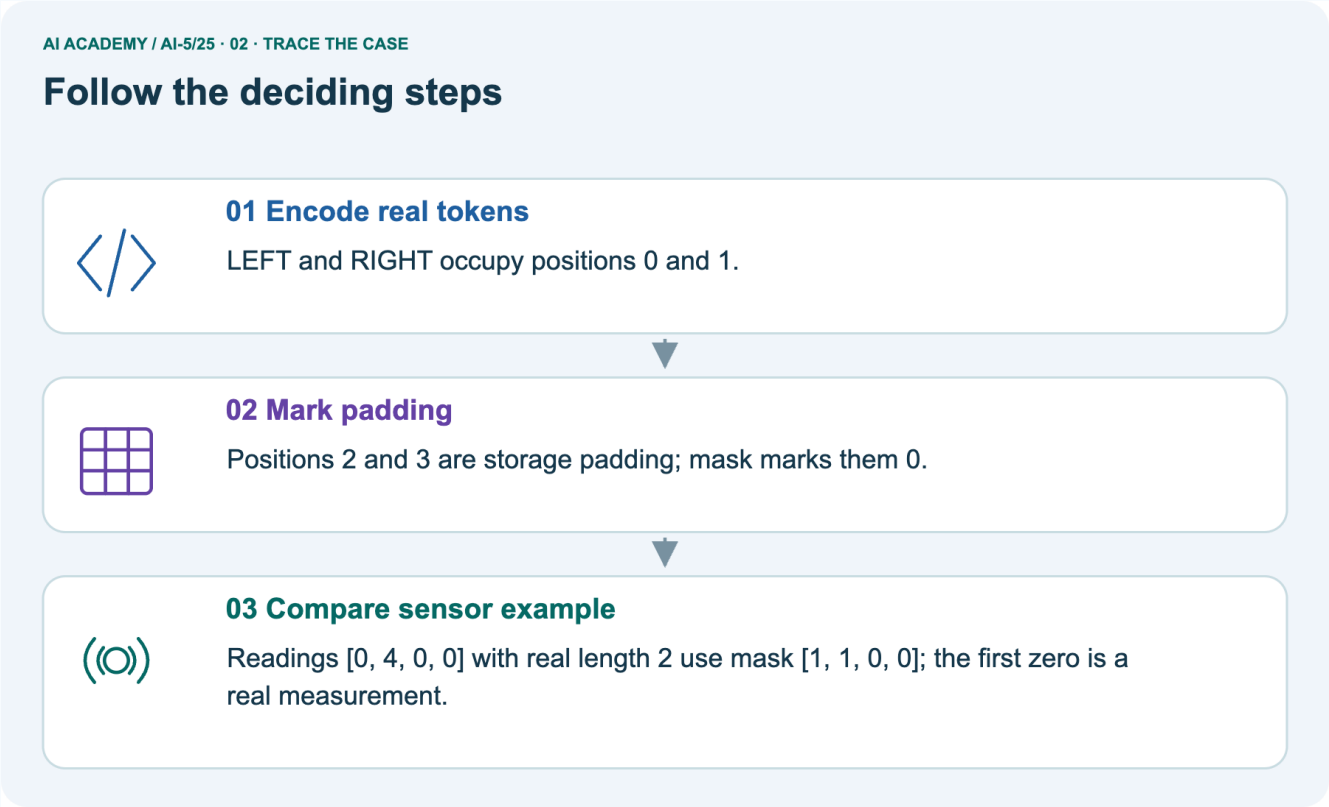
Follow the labelled picture with the matching explanation. Open the vector diagram

Distinguish **padding** positions from meaningful zero sensor readings. Supplied facts: Toy tokens PAD 0, UNK 1, LEFT 2, RIGHT 3; stored length 4. **Sequence** LEFT RIGHT encodes [2, 3, 0, 0], **mask** [1, 1, 0, 0].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Encode real tokens: LEFT and RIGHT occupy positions 0 and 1. → Mark **padding**: Positions 2 and 3 are storage **padding**; **mask** marks them 0. → Compare sensor example: Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.

Point to each numbered stage and explain which recorded **input** it uses.

03. A padded log still has two real events

AI ACADEMY / AI-5/25 · 03 · INSPECT THE LESSON MODEL

A padded log still has two real events

Stored position	Token ID	Mask
0	2 meaning GO	1 real
1	4 meaning STOP	1 real
2	0 meaning PAD	0 placeholder
3	0 meaning PAD	0 placeholder

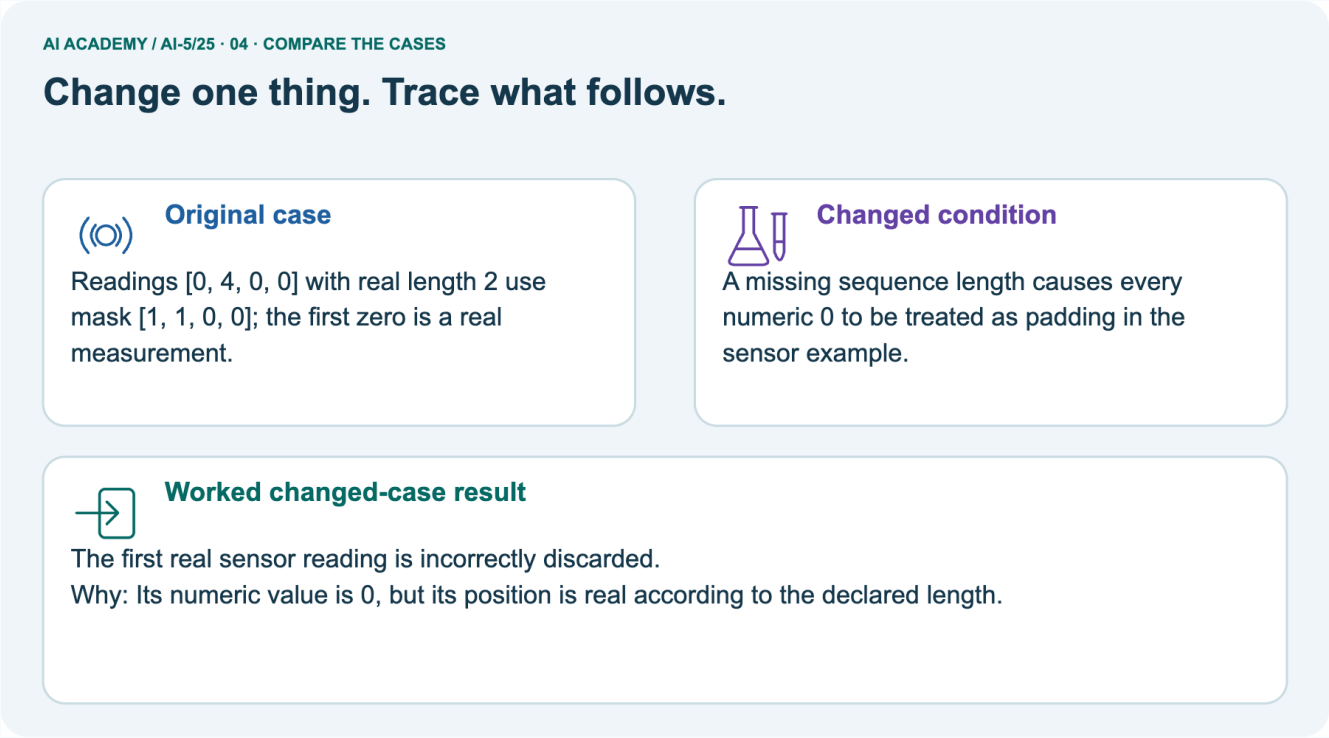
The final stored slot is padding. For this right-padded contiguous mask, the real length is the mask sum, and the final real event is at length minus one. A different masking layout would need a different indexing rule.

Follow the labelled picture with the matching explanation. Open the vector diagram

A padded log still has two real events. 0 | 2 meaning GO | 1 real. 1 | 4 meaning STOP | 1 real. 2 | 0 meaning PAD | 0 placeholder. 3 | 0 meaning PAD | 0 placeholder

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

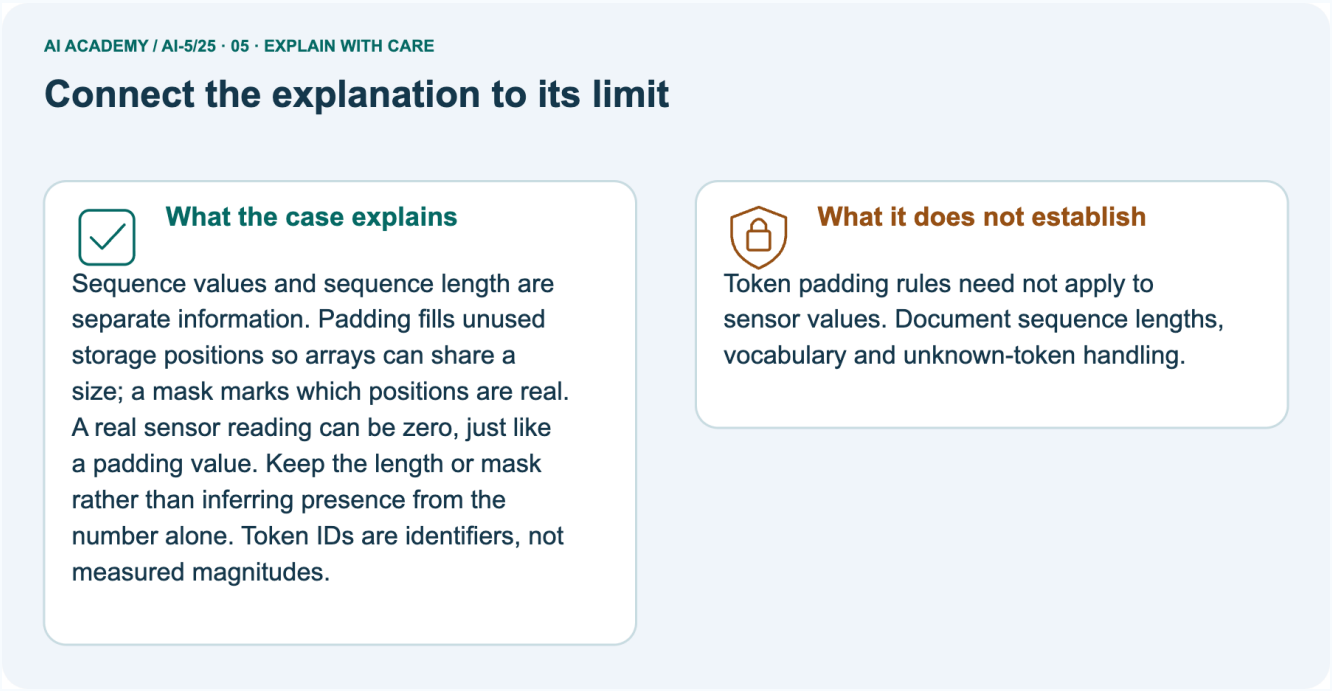
Original case: Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.

Changed condition: A missing **sequence** length causes every numeric 0 to be treated as **padding** in the sensor example. Expected result: The first real sensor reading is incorrectly discarded.

Reason: Its numeric value is 0, but its position is real according to the declared length.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

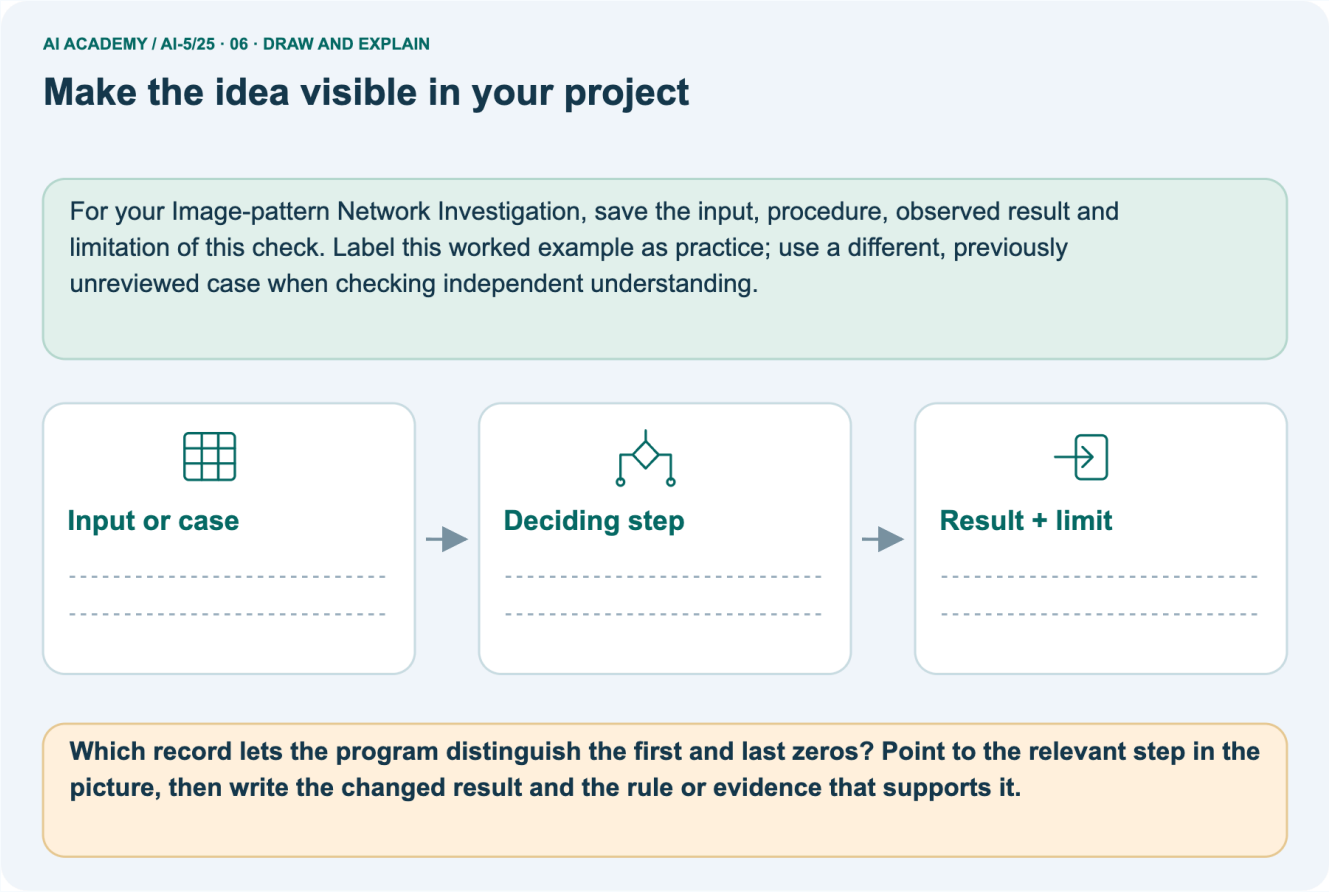
Sequence values and **sequence** length are separate information. **Padding** fills unused storage positions so arrays can share a size; a **mask** marks which positions are real.

A real sensor reading can be zero, just like a **padding** value. Keep the length or **mask** rather than inferring presence from the number alone. **Token ID**s are identifiers, not measured magnitudes.

Limit: Token **padding** rules need not apply to sensor values. Document **sequence** lengths, vocabulary and unknown-token handling.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Sequence representations: the case

02. Follow the deciding steps

03. A padded log still has two real events

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.